OOD Quantification → Category-Shift Adaptation

Feature- & sample-level weights with a soft cold-start gate

AS-DTA / CFPerm

Abstract

Message. Category shift is an OOD event on the live stream. We *quantify* the OOD footprint once (RF-domain VIMP + membership), then *adapt* a CE head with two cheap weights via `partial_fit`—no new backbone, tiny compute. v2 uses a **soft cold-start gate** decay = $(1-n_{\text{idx}}/N_{\text{cap}})_+$ (N_{cap}=200): empower the scarce regime, taper to uniform as the new category warms up. On text multiclass boards (FULL LIVE holdout), early overall-acc lift is clear; past the cap, soft-gated weights stay above uniform without needing an EWMA schedule.

1 OOD quantification → category-shift adapt

Problem. REF covers known categories; LIVE mixes known support with novel labels (category shift). Question: *where is the OOD mass, and how do we give it CE budget cheaply?*

Quantify. Score batch membership $W\in\{0,1\}$ (REF vs LIVE) with a shallow RF. Importances $v \in \Delta^{p-1}$ are the OOD coordinates. On the top-K of v, each live point's z-distance to REF yields membership $m_i \in [0,1]$ ("new-category-like"). That is the sensor; novel labels only fuse into m_i at adapt time.

Adapt (two orthogonal weights).

- **Feature-level:** $w_j^{\text{feat}} = 1 + \alpha_{\text{eff}}\,\tilde{v}_j$, $x_{ij} \leftarrow x_{ij}\,w_j^{\text{feat}}$ (stretch OOD axes; α_{eff} tapered by mean decay).
- **Sample-level:** excess membership weight × per-row decay, passed as `sample_weight` into `partial_fit`.

Same OOD quantification; two actuators on the SGD update. Implementation: one RF pass + a few linear CE `partial_fit` epochs.

Soft cold-start gate (v2). With within-category arrival index $n_{\text{idx}}(i)$,

$$\text{decay}_i = \big(1 - n_{\text{idx}}(i)/N_{\text{cap}}\big)_+, \qquad N_{\text{cap}} = 200, \tag{1}$$

on novel rows (known stay at decay 0). Boost excess is multiplied by decay_i; feature stretch uses mean decay. Hard cut and always-on are ablations: soft is the default—empower early, taper, get out of the way.

Cost. No architecture change, no replay redesign, no EWMA. Compute ≈ one small RF + linear CE `partial_fit`.

2 Learning curves (FULL LIVE holdout)

KPI = overall accuracy and mean novel OVA-AUC on a fixed full LIVE holdout. Budget = novel samples / category. Dashed line = N_{cap}.

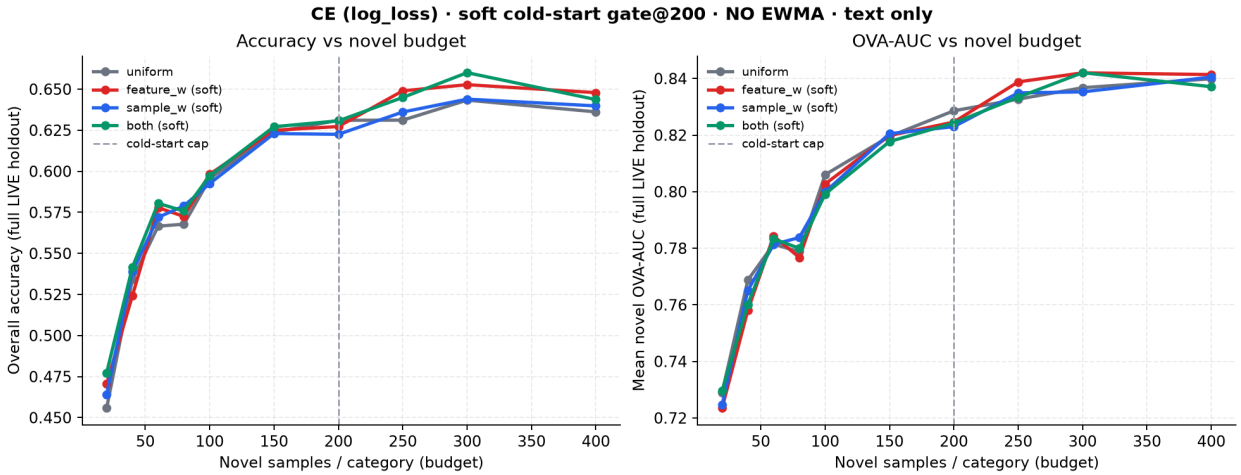


Figure 1: Soft-gate schemes vs uniform. Left: overall accuracy. Right: mean novel OVA-AUC. Early regime: weighted schemes lead (empowerment). Past 200: soft curves stay near / above uniform.

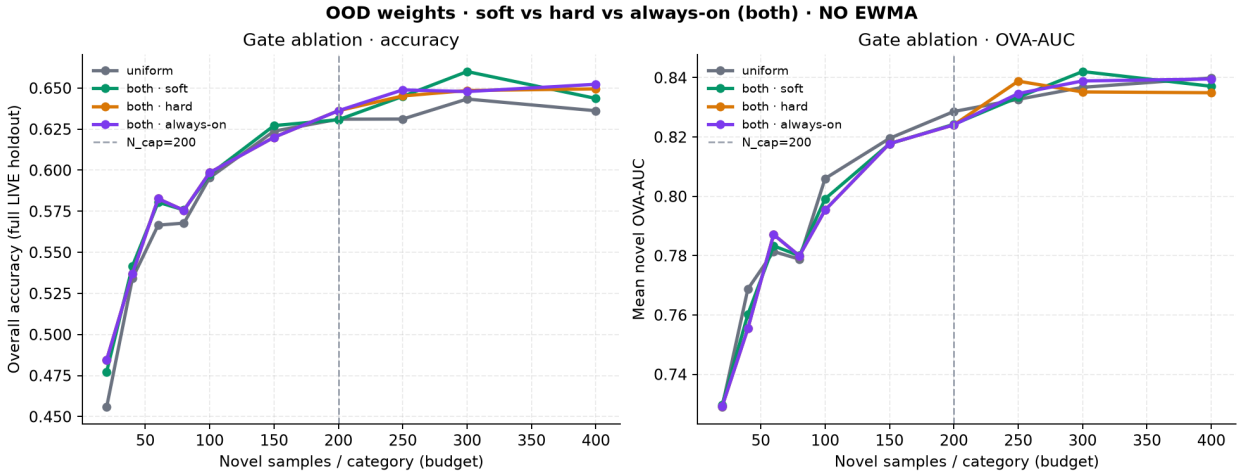


Figure 2: Gate ablation on the `both` channel: soft (default), hard cut, always-on. Soft matches hard early and avoids a brittle cliff; always-on can keep a small late edge but keeps boosting warm categories.

Reading the figures.

- **Empowerment works.** At b=20/cat, mean overall acc 0.456 → **0.477** under soft `both` (Δ+0.021).
- **Soft gate is enough.** Endpoint mean acc: uniform 0.636, soft `both` 0.644 (Δ+0.007), soft `feature` **0.648** (Δ+0.012).

- **OVA-AUC stays flat.** Ranking barely moves; the operational gain is overall acc under CE.

3 Endpoint (past the gate)

	uniform	feature (soft)	sample (soft)	both (soft)
mean overall acc	0.636	**0.648**	0.640	0.644
mean Δacc vs U	—	+0.012	+0.004	+0.007

Feature-level helps most when a few token dimensions carry the shift (`cnae9`, Amazon); soft `both` leads when the new category is scarcest (Fig. 1, left of the dashed line).

4 Takeaway

1. **Quantify OOD, then adapt.** RF-domain VIMP + membership is the category-shift sensor; feature / sample weights are the actuator.
2. **Keep the head; use `partial_fit`.** One linear CE update with `sample_weight` (and an optional column scale) ships beside an existing online pipeline.
3. **Empower with a soft gate.** decay $= (1 - n_{\text{idx}}/200)_+$: clear early lift, taper when warm, tiny compute—no EWMA.